



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Name: Muhammad Musawar Ali Shah

Roll No. 24P-0619

Dept: BS-CS

Section: BCS-3D

Subject: COAL LAB

Submitted To: Sir Hasaan Shaukat

Lab 6

Question 1:

Code	Explanation
<pre>[org 0x0100] mov al, 6 mov bl, 5 cmp al, bl je equal jne notEqual equal: mov byte [num1], 1 notEqual: mov byte [num1], 0 mov ax, 0x4c00 int 0x21 num1: db 0</pre>	▶ Stored the value 6 in al register ▶ Stored value 5 in bl register ▶ Compare al and bl ▶ If al = bl the store 1 in the result ▶ else if al != bl then store 0 in the result

Question 2:

Code	Explanation
<pre>[org 0x0100] mov ax, -3 cmp ax, 0 je isZero jl isNegative isPositive: mov byte [check], 1 jmp done isNegative: mov byte [check], 2 jmp done isZero: mov byte [check], 0 done: mov ax, 0x4c00 int 0x21</pre>	▶ Store some value in ax register to check if it is negative, positive or zero ▶ If the value is positive, jump to isPositive: ▶ Else if the value is negative, jump to isNegative. ▶ Else jump to isZero: ▶ If the value is positive, store 1 in the result ▶ else if the value is negative, store 2 in the result ▶ else store 0 in the result

check: db 0	
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Question 3:

Code	Explanation
<pre>[org 0x0100] mov cx, 5 start: dec cl jnz start mov ax, 0x4c00 int 0x21</pre>	▶ Store 5 in cx register. ▶ Decrease value of cx register until it gets 0 result. ▶ End the program